

Laboratório 6

```
library(DBI)
library(RSQLite)
conn <- dbConnect(SQLite(), "disco.db")
```

3. Liste as tabelas existentes no banco de dados.

```
dbListTables(conn)
```

```
[1] "albums"          "artists"          "customers"         "employees"
[5] "genres"          "invoice_items"    "invoices"          "media_types"
[9] "playlist_track"  "playlists"        "sqlite_sequence"   "sqlite_stat1"
[13] "tracks"
```

4. Identifique os nomes de todas as colunas existentes na tabela customers.

```
dbListFields(conn, 'customers')
```

```
[1] "CustomerId"  "FirstName"  "LastName"  "Company"  "Address"
[6] "City"        "State"     "Country"   "PostalCode" "Phone"
[11] "Fax"         "Email"     "SupportRepId"
```

5. Utilizando apenas SQLite, com o apoio do comando dbGetQuery, identifique quantos clientes estão atualmente cadastrados neste banco de dados.

```
dbGetQuery(conn, "SELECT COUNT(*) AS total_clientes
FROM customers
")
```

```
total_clientes
1              59
```

6. Utilizando apenas SQLite, identifique o número de países diferentes em que moram os clientes encontrados acima.

```
dbGetQuery(conn, "  
  SELECT COUNT(DISTINCT Country) AS total_paises  
  FROM customers  
")
```

```
total_paises  
1           24
```

7. Utilizando apenas SQLite, quantos clientes existem por país? A tabela resultante deve conter o nome do país e a respectiva contagem, além de ser ordenada de maneira decrescente pela referida contagem.

```
dbGetQuery(conn, "  
  SELECT Country, COUNT(*) AS total_clientes  
  FROM customers  
  GROUP BY Country  
  ORDER BY total_clientes DESC  
")
```

	Country	total_clientes
1	USA	13
2	Canada	8
3	France	5
4	Brazil	5
5	Germany	4
6	United Kingdom	3
7	Portugal	2
8	India	2
9	Czech Republic	2
10	Sweden	1
11	Spain	1
12	Poland	1
13	Norway	1
14	Netherlands	1
15	Italy	1
16	Ireland	1
17	Hungary	1
18	Finland	1
19	Denmark	1

20	Chile	1
21	Belgium	1
22	Austria	1
23	Australia	1
24	Argentina	1

8. Quais são os 5 países com mais clientes registrados? Use apenas SQLite

```
dbGetQuery(conn, "
SELECT Country, COUNT(*) AS total_clientes
FROM customers
GROUP BY Country
ORDER BY total_clientes DESC
LIMIT 5
")
```

	Country	total_clientes
1	USA	13
2	Canada	8
3	France	5
4	Brazil	5
5	Germany	4

9. Quais são os países registrados que possuem apenas 6 letras no nome?

```
dbGetQuery(conn, "
SELECT DISTINCT Country
FROM customers
WHERE Country LIKE '_____'
")
```

	Country
1	Brazil
2	Canada
3	Norway
4	France
5	Poland
6	Sweden

10. Quais foram as músicas compradas por clientes brasileiros?

```

dbGetQuery(conn, "
  SELECT DISTINCT tracks.Name AS musica
  FROM tracks
  INNER JOIN invoice_items
    ON tracks.TrackId = invoice_items.TrackId
  INNER JOIN invoices
    ON invoice_items.InvoiceId = invoices.InvoiceId
  INNER JOIN customers
    ON invoices.CustomerId = customers.CustomerId
  WHERE customers.Country = 'Brazil'
  ORDER BY musica
")

```

	musica
1	1/2 Full
2	2 Minutes To Midnight
3	A Cor Do Sol
4	A Cura
5	A Menina Dança
6	Abraham, Martin And John
7	Aces High
8	Admirável Gado Novo
9	All Along The Watchtower
10	All I Want Is You
11	Aloha
12	Amor De Muito
13	Animal
14	Any Colour You Like
15	Aos Leões
16	Babyface
17	Back off Bitch
18	Bad
19	Bad Seed
20	Banditismo Por Uma Questa
21	Be Good Johnny
22	Be Mine
23	Big Wave
24	Binky The Doormat
25	Bittersweet Me
26	Black
27	Black Light Syndrome

28	Bora-
Bora	
29	Bossa
30	Burden In My Hand
31	Calling Dr. Love
32	Carolina Hard-
Core Ecstasy	
33	Casa
34	Caso Você Queira Saber
35	Child In Time
36	Cinema Mudo
37	Cold Gin
38	Coma
39	Communication Breakdown(2)
40	Cristina Nº 2
41	Cropduster
42	Cry For Love
43	Cérebro Eletrônico
44	D'Yer Mak'er
45	Damage Inc.
46	Dance
47	Demorou!
48	Dissident
49	Don't Cry (Original)
50	Don't Look Back
51	Doutor
52	Down Under
53	Down by the Sea
54	Drifter
55	Duelists
56	Dust N' Bones
57	Electrolite
58	Eu Amo Você
59	Everything I Need
60	Experiment In Terra
61	Eye
62	Fantasia On Greensleeves
63	Fast As a Shark
64	Flower
65	Flying High Again
66	Fool In The Rain
67	Garden of Eden
68	Get Down, Make Love

69	Get Up
70	Girl From A Pawnshop
71	Go Down
72	Green Disease
73	Grito De Alerta
74	Heart Of Lothian: Wide Boy / Curtain Call
75	Hell Ain't A Bad Place To Be
76	Help Yourself
77	Home Sweet Home
78	I Belong To You
79	I Feel Good (I Got You) - Sossego
80	In Bloom
81	In The Evening
82	Interlude Zumbi
83	Into The Light
84	Intro
85	Ipiranga 2001
86	Is This Love (Live)
87	It's a Mistake
88	Jesus Christ Pose
89	Karelia Suite, Op.11: 2. Ballade (Tempo Di Menuetto)
90	Knocking At Your Back Door
91	Lay Down Sally
92	Leash
93	Leper Messiah
94	Like A Song...
95	Lindo Lago Do Amor
96	Linha Do Equador
97	Live and Let Die
98	Lixo Do Mangue
99	Losfer Words
100	Mama, I'm Coming Home
101	Maracatu De Tiro Certeiro
102	Mateus Enter
103	Maybe I'm A Leo
104	Mensagen De Amor (2000)
105	Meu Caro Amigo
106	Meu Erro
107	Midnight From The Inside Out
108	Mis Penas Lloraba Yo (Ao Vivo) Soy Gitano (Tangos)
109	Morena De Angola
110	Most High
111	Mundaréu

112	Music for the Funeral of Queen Mary: VI. "Thou Knowest, Lord, the Secrets of Our Hearts"
113	Nega Do Cabelo Duro
114	No No No
115	No Quarter
116	Nossa Gente (Avisa Là)
117	Nosso Adeus
118	Não Vou Ficar
119	O Descobridor Dos Sete Mares
120	O Leaozinho
121	O Que Me Importa
122	Oceans
123	On A Plain
124	Pais E Filhos
125	Paranoid
126	Pick Myself Up
127	Plot 180
128	Podes Crer
129	Ponto De Interrogação
130	Powerslave
131	Praise
132	Red Light
133	Redundant
134	Refavela (Live)
135	Reggae Tchan
136	Revolta Olodum
137	Right Next Door to Hell
138	Rios Pontes & Overdrives
139	Rock And Roll Is Dead
140	Rocket Queen
141	Saber Amar
142	Samba Do Lado
143	Sanctuary
144	Saudade Dos Aviões Da Panair (Conversando No Bar)
145	Scentless Apprentice
146	Selvagem
147	Sereia
148	Será
149	Será Que Vai Chover?
150	Seven Seas Of Rhye
151	Shakes and Ladders
152	Shout It Out Loud
153	Sincero Breu
154	Slither

155		Snowballed
156		So Fast, So Numb
157		Sometimes Salvation
158		Soul Singing
159		Stand Inside Your Love
160		Stir It Up (Live)
161	String Quartet No. 12 in C Minor, D. 703 "Quartettsatz": II. Andante - Allegro assai	
162		Strutter
163	Suite No. 3 in D, BWV 1068: III. Gavotte I & II	
164		Sweetest Thing
165	Symphony No. 2, Op. 16 - "The Four Temperaments": II. Allegro Comodo e Flegmatico	
166		Take the Celestra
167		The Day I Tried To Live
168		The Great Gig In The Sky
169		The Memory Remains
170		The Unforgiven II
171		The Unwritten Law
172		Think About You
173		Title Song
174		Top Top
175		TriboTchan
176		Um Satélite Na Cabeça
177		Untitled
178		Vai Passar
179		Vai Valer
180		Vamo Batê Lata
181		Vavoom : Ted The Mechanic
182		Voce Nao Entende Nada - Cotidiano
183		Wanted Dread And Alive
184		Why Go
185		X-
9 2001		
186		You're My Best Friend
187		You've Been A Long Time Coming
188		Your Blue Room
189		Água de Beber

```
dbDisconnect(conn)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

[1] "2026-09-21 07:04:29 -03"